

Solution Requirements

Date	13 July 2026
Team ID	
Project Name	SkillEcho — Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Functional Requirements (FR) define specific behaviours, functions, or features of the system (what the system should do). Non-Functional Requirements (NFR) specify the criteria that judge the operation of the system rather than specific behaviours (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
FR-01	Authentication	System shall allow new users to register with name, email, password, and role (student/admin) via POST /auth/register.	High
FR-02	Authentication	System shall authenticate users via email and SHA-256 hashed password comparison (POST /auth/login).	High
FR-03	Authorization levels	System shall enforce role-based access control — students access evaluation features; admins access concept management and all-user results.	High
FR-04	Reference Concept Mgmt	Admins shall create reference concepts with title, description text, and optional PDF upload (POST /admin/concepts).	High
FR-05	Reference Concept Mgmt	Admins shall update existing concepts (title, text, PDF replacement/clearing) via PUT /admin/concepts/{id}.	Medium
FR-06	Reference Concept Mgmt	Admins shall delete concepts, removing the concept and associated PDF from disk.	Medium
FR-07	Reference Concept Mgmt	Students shall view the list of all available concepts via GET /concepts.	High
FR-08	Reference Concept Mgmt	Students shall download the reference PDF for a concept via GET /concepts/{id}/pdf.	Medium
FR-09	Reference Concept Mgmt	System shall seed 5 initial reference concepts (ML, DL, NLP, RL, Neural Networks) on first boot if the database is empty.	Low
FR-10	External interfaces	Students shall upload a WAV audio file of their spoken explanation via the Streamlit UI as multipart form data.	High
FR-11	Transactions processing	Uploaded audio files shall be persisted to data/uploaded_audio/ with UUID-prefixed filenames to prevent collisions.	High
FR-12	Transactions processing	System shall transcribe uploaded audio to text using OpenAI Whisper (tiny model, English, fp16=False).	High
FR-13	Transactions processing	The Whisper model shall be loaded once per process lifetime and cached at module level.	Medium
FR-14	Transactions processing	System shall extract acoustic signal features: RMS energy, zero-crossing rate, pause/silence ratio, and duration.	High
FR-15	Transactions processing	System shall compute filler-word statistics (count, ratio, breakdown) for um/uh/like/so/actually/basically.	High
FR-16	Business rules	System shall perform a Topic Guardrail check comparing core vocabulary overlap between transcript and reference text; on mismatch, heavy semantic models are skipped.	High

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FR-17	Transactions processing	System shall compute Bi-Encoder semantic similarity using all-MiniLM-L6-v2 (cosine similarity of L2-normalised embeddings).	High
FR-18	Transactions processing	System shall compute Cross-Encoder semantic accuracy using ms-marco-MiniLM-L-6-v2 (sigmoid-scaled logits).	High
FR-19	Business rules	System shall compute a deterministic overall score (0–100) via a weighted matrix: Bi-Encoder 25, Cross-Encoder 25, Filler 20, Pause 15, RMS 15.	High
FR-20	Business rules	System shall classify scores into Strong (≥ 80), Moderate (≥ 50), Poor (< 50), or Topic Mismatch (0) tiers.	High
FR-21	Reporting	All evaluation artefacts shall be persisted across 10 normalised PostgreSQL tables (users, audio_files, transcripts, filler_word_stats, reference_concepts, semantic_similarities, audio_features, evaluation_results, sessions, reports).	High
FR-22	Business rules	Audio file records shall update to status="completed" with duration_sec on success, or status="failed" on error.	Medium
FR-23	Reporting	System shall generate a professional in-memory PDF report (ReportLab Platypus) with header, reference text, transcript, waveform chart, metrics table, and footer.	High
FR-24	Reporting	The PDF metrics table shall display Topic Guardrail status, Bi-Encoder %, Cross-Encoder %, Filler ratio, Pause ratio, RMS energy, Overall Score, and colour-coded Understanding Level.	High
FR-25	Reporting	Students shall view their evaluation history with full metric detail via GET /users/{user_id}/history.	Medium
FR-26	Reporting	Admins shall view all evaluation results across all students via GET /admin/results.	Medium
FR-27	Other	Admins shall view all registered users, ordered by created_at DESC, via GET /admin/users.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
NFR-01	Performance & Speed	End-to-end evaluation pipeline (upload → scorecard) shall complete within 30 seconds for a 60-second audio clip.	≤ 30 s response time
NFR-02	Scalability	Application shall operate within Streamlit Community Cloud's resource ceiling using aggressive garbage collection and the Whisper tiny model.	< 1 GB peak RAM
NFR-03	Security & Data Privacy	User passwords shall be stored as SHA-256 hex digests; plaintext passwords are never persisted or logged.	SHA-256 hashing, verified by code review
NFR-04	Security & Data Privacy	Database credentials shall be stored in environment variables (.env), never hardcoded in source.	.env config, excluded via .gitignore
NFR-05	Reliability & Availability	Failed evaluations shall not corrupt the database; all DB operations use context-managed sessions with automatic rollback.	Transactional safety (commit/rollback/closure)
NFR-06	Reliability & Availability	Platform shall be deployable on Streamlit Community Cloud with zero manual infrastructure provisioning.	Cloud-ready repo (requirements.txt, packages.txt, config.toml)
NFR-07	Scalability	Database layer shall support PostgreSQL (prod) and SQLite (dev) via a configurable DATABASE_URL.	Dual-DB support, auto-detected by URL prefix
NFR-08	Usability & Accessibility	Codebase shall follow Domain-Driven Design with clear src/frontend and src/backend separation and modular services.	Directory structure audit

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NFR-09	Reliability & Availability	Whisper model shall run with fp16=False for broad CPU compatibility across cloud environments.	CPU-only inference verified
NFR-10	Usability & Accessibility	Streamlit frontend shall show loading spinners during AI processing to maintain engagement during the 15–25s pipeline.	UX feedback, verified by visual inspection
NFR-11	Security & Data Privacy	PostgreSQL schema shall enforce foreign-key constraints with ON DELETE CASCADE to prevent orphaned records.	Referential integrity in all models.py ForeignKeys
NFR-12	Other	All API requests shall be logged with structured logging (timestamp, level, module, message).	Structured logs via logging.basicConfig()